

ASSIGNMENT-II

Deadline: 05/04/2026

Q.1 A router has the following (CIDR) entries in its routing table:

Address/mask	Next hop
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?

- a. 135.46.63.10
- b. 135.46.57.14
- c. 135.46.52.2
- d. 192.53.40.7
- e. 192.53.56.7

Q.2 A large number of consecutive IP addresses are available starting at 194.32.0.0. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.

Q.3 Explain the IPv6 addressing and working of tunneling.

Q.4 Discuss the Multicast Routing and Broadcast Routing with the help of example.

Q.5 Write the short notes on RIP, OSPF, and BGP routing protocols.

Q.6 Define ARP, RARP, BOOTP, and DHCP protocol.